Z-Spin Cosmology — Proprietary Terminology Glossary

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> **Scope:** This glossary catalogs terms coined within the Z-Spin Cosmology framework that are not standard terminology in mainstream physics or mathematics. Terms borrowed from established physics (e.g., "Horndeski," "Goldstone," "seesaw") are excluded unless Z-Spin introduces a fundamentally new meaning.
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> **v1.0 changelog:** 6 new entries (Fractal Symmetry, Cross-Coupling Theorem, Axion-Free Strong CP, Seam-Charge Projection, Hubble Holonomy Mapping, Z-Telomere Collapse). Mirror Universe entry updated. Total: 88 entries across 13 categories.

I. Framework & Architecture

Z-Spin

The overarching theoretical framework name. Refers to the role of spin as the dynamical operator mediating between the X (macroscopic) and Y (microscopic) sectors through the Planck-scale Z sector, combined with the i-tetration self-referential rotation that governs the Z-sector dynamics.

Z-Spin Cosmology

The complete scalar-tensor cosmological theory built on the Z-Spin framework. Defined by a single geometric constant A = 35/437, with zero free parameters beyond standard physics inputs. The action: $S = \int d^4x \sqrt{(-g)} [\frac{1}{2}M^2_P(1+A|\Phi|^2)R - \frac{1}{2}M^2_P|\partial\Phi|^2 - V(\Phi)] + S_m$.

X Sector

The macroscopic sector: 3 algebraic spatial dimensions corresponding to the "particle + space" layer. Spatial freedom dominates; time is "point-like" (confined). Associated with the cube–octahedron Platonic dual pair. X = 3.

Y Sector

The microscopic sector: 6 dimensions encoding gauge symmetries and quantum information. The "wave + time" layer, where temporal freedom dominates and space is "point-like." Associated with the dodecahedron–icosahedron Platonic dual pair. Y = 6.

Z Sector

The 2-dimensional Planck-scale boundary surface mediating between X and Y sectors. The "entanglement interface" where time-points and space-points exchange information. Associated with the self-dual tetrahedron pair. $Z = 2$.

Q = 11 Register (Slot Register)

The total dimensionality of the Z-Spin geometric universe: $Q = X + Y + Z = 3 + 6 + 2 = 11$. Functions as an 11-slot quantum register in the information-theoretic formulation. Source: ZS-F5.

G = 12 (Gauge Algebra Dimension)

$G = Q + 1 = 12$, the total gauge algebra dimension. Connected to Q through the number-theoretic identity $MUB(Q) = Q + 1$ for prime Q (mutually unbiased bases). Encodes the Standard Model gauge structure $\dim(su(3) \oplus su(2) \oplus u(1)) = 8 + 3 + 1 = 12$.

Z-EFT (Z-Completed Effective Field Theory)

The $U(1)_Z$-completed Z-Spin effective field theory employing the complex scalar field Φ as the fundamental dynamical variable. Source: ZS-F1.

II. Core Constants & Mechanisms

Geometric Impedance (A = 35/437)

The resistance ratio arising during Planck-scale information exchange between the macroscopic universe (X) and microscopic universe (Y). The geometric resistance encountered when Y-sector (6D) physical quantities are converted through the Z-sector (2D boundary) into the X-sector (3D sphere). Fixed by the $O_h \times I_h$ polyhedral defect structure; not a fitted parameter. $A = 35/437 \approx 0.080092$. Source: ZS-F2.

i-Tetration Fixed Point (z*)

The unique attractive fixed point of $z = i^z$, found at $z^* = 0.4383 + 0.3606i$ (Lambert W principal branch $k_W = 0$). Central to Z-Spin dynamics as the self-referential rotation operator of the Z sector. All other branches ($k_W \neq 0$) are repulsive. Source: ZS-F3.

η_{topo} (Topological Fill Fraction)

$\eta_{topo} \equiv |z^*|^2 = 0.3221$. The squared modulus of the i-tetration fixed point, interpreted as the topological fill fraction of the Z-sector information channel. Matches Ω_m to 0.06% via an independent mathematical route from sector counting.

Stability Threshold (S)

$S \equiv |z^*| \cdot (\pi/2) = 0.8915$. The fraction of the stability budget consumed by the i-tetration fixed point. Determines the three-generation limit and confinement threshold.

Holonomy Accumulation (exp(A))

The exponential of the geometric impedance, exp(A) = exp(35/437) = 1.0834. Arises from path-ordered parallel transport (Wilson loop of the scalar connection) on the polyhedral manifold. Governs the H_0 local/CMB ratio. Source: ZS-F5.

Hubble Holonomy Mapping

The specific application of holonomy accumulation to the Hubble tension: H_0^local / H_0^CMB = exp(A) = 1.0834. Demonstrates that the discrepancy between local distance-ladder measurements (SH0ES: 73.04 ± 1.04 km/s/Mpc) and CMB inference (Planck: 67.36 ± 0.54 km/s/Mpc) is not a measurement error but a geometric parallel-transport effect — a path-ordered exponential around the polyhedral defect lattice relating the Jordan frame (local measurements) and Einstein frame (CMB inference). Prediction: H_0^pred = 72.98 km/s/Mpc (0.06σ from SH0ES). Source: ZS-F3, ZS-F4 §6, ZS-U5 §3.1.

$2e^A$ Duality Identity

The scale-invariant identity Ω_Λ/Ω_m = m_d/m_u = $2e^A$ = 2.1668. Connects cosmic scales (Gpc) to quark scales (fm) through a single expression spanning 40 orders of magnitude. Source: ZS-A1, ZS-A5.

δ-Uniqueness Theorem

Proves that A = 35/437 is the unique value satisfying six axioms (A0–A6: boundedness, continuity, normalization, monotonicity, holonomy composition, linearization). The composition axiom A5 (Einstein addition) is the decisive constraint: the unique way to combine holonomy additivity and 1-loop log-determinant in a bounded observable. Source: ZS-F5 §3.

Duality-Deviation Invariant δ(P)

Defined as δ(P) = |F−V|/(F+V) for polyhedron P. Measures deviation from Platonic duality (F ↔ V). A = 35/437 is uniquely determined from the defect structure of O_h (octahedral group) and I_h (icosahedral group). Source: ZS-F2 §2.2.

III. Fields & Dark Sector

Z-Field (Φ)

The fundamental complex scalar field of Z-Spin Cosmology: Φ(x) = |Φ|e^{iθ}. Its radial mode |Φ| (the ε-field) is kinetically frozen at Planck mass; its angular mode θ (Goldstone) is massless and generates dark sector phenomenology.

ε-Field (Radial Mode)

The radial component of the Z-field, with mass m_ρ ~ O(M_P) ≈ 1.34 M_P. Frozen at the cosmological attractor ε → 1. Generates the universal G_eff = G/(1+A) rescaling. Compton wavelength λ_C ≈ 6×10^{-35} m — no propagating fifth force at any astrophysical scale.

θ-Field (Goldstone Angular Mode)

The massless Goldstone angular mode of the Z-field. Satisfies $\Box\theta = 0$ exactly (Goldstone theorem). Its gradient energy produces the ε-Halo (dark matter replacement) and its vacuum energy produces the ε-Drive (dark energy replacement).

ε-Halo

Z-Spin replacement for "Dark Matter." The gradient energy density of the Goldstone θ-mode: $\rho \propto 1/r^2$ (isothermal halo). Produces flat rotation curves with zero dark matter particles. X-sector origin (galactic θ-gradient). Source: ZS-F4, ZS-A1, ZS-A2.

ε-Drive

Z-Spin replacement for "Dark Energy." The Y-sector attractor mode producing the cosmological constant through V_0 (the vacuum energy at the $\varepsilon = 1$ attractor). Same Z-field action as ε-Halo — not a separate component. Source: ZS-F4, ZS-A2.

ε-Higgs Portal

The effective coupling between the ε-field and the Higgs doublet H, arising from the conformal factor $\Omega^2 = 1 + A\varepsilon^2 + 2\xi(H^\dagger H)/M^2_P$. Effective coupling $\xi_{eff} \sim A^2/(16\pi^2) \approx 4 \times 10^{-5}$. Preserves Higgs potential shape while all ε–H interactions remain perturbatively small. Source: ZS-S4.

Z-Bias Field

The field describing the microscopic origin of the i-tetration fixed point $z^* = i^{z^*}$. The fixed point of the Z-sector transfer map $T(z) = i^z$ serves as the vacuum expectation value of the Z-Bias field. Source: ZS-M1.

θ-Halo

The isothermal halo created by gradient energy of the Goldstone θ-mode in galactic disks. Solution of the 2D Laplace equation $\Box\theta = 0$: $\theta(r) = \ln(r/r_s)/L \rightarrow \rho \propto 1/r^2$. Extended to elliptical galaxies via the Vortex Glass Network. Source: ZS-A1 §2.

IV. Structural Mechanisms

Seam Involution ($\hat{W}$)

The $\mathbb{Z}_2$ topological operator satisfying $\hat{W}^2 = I$ that divides the universe into trivial and non-trivial sectors. Exchanges $X \leftrightarrow Y$ sector projections: $\hat{W}(\Psi_X, \Psi_Y)^T = (\Psi_Y, \Psi_X)^T$. Serves as the criterion distinguishing baryons and dark matter; enforces B−L conservation in the Standard Model projection. Source: ZS-U3.

Seam Gate (J)

The channel-level antiunitary conjugation operator implementing the seam involution on quantum states: $J|j\rangle = |Q-1-j\rangle$ for the $Q = 11$ register (i.e., $J|j\rangle = |10-j\rangle$). Satisfies $J^2 = I$ with eigenspace dimensions $\dim(E_+) = 6$, $\dim(E_-) = 5$. Source: ZS-M3, ZS-A4.

Seam Witness (u_seam)

The co-primary measurable endpoint for testing seam symmetry: $u_seam = \|(J\otimes J)C_\Lambda(J\otimes J) - C_\Lambda^T\|_F / \|C_\Lambda\|_F$, where C_Λ is the Choi state of the effective horizon channel Λ. Under the seam constraint, $u_seam = 0$ exactly. Source: ZS-A4.

κ = 4 Witness

The unique nontrivial gauge-preserving $\mathbb{Z}_2$ automorphism on su(3), where κ counts the number of ladder generators whose sign is flipped. Witness: $U = \mathrm{diag}(1,-1,-1)$. Selected by the gauge preservation condition $\kappa \leq r = 4$ (ZS-F5 bound). Source: ZS-F5.

Z-Anchor

The topological boundary condition at the center of topological vortices (such as SMBH event horizons) where the Z-field magnitude vanishes ($|\Phi| = 0$). In U(1) completion, $\pi_1(U(1)) = \mathbb{Z}$ winding forces $|\Phi| \to 0$ at vortex cores. The horizon is the Z-sector boundary where X ↔ Y sector exchange occurs. Source: ZS-A3, ZS-A1 v1.0.

Z-Telomere

The phase-drift/accumulation mechanism determining the topological life-cycle of the universe. Each primitive cell accumulates a phase drift $\delta\varphi = A$, and when this accumulation reaches a critical threshold, a phase transition or tunneling event occurs. Micro-period $T_micro = 2\pi/A \approx 78.45$ Planck cycles. Telomere collapse (topological holonomy winding-change, not CdL tunneling) requires $S_tunnel = 5$ seam flips [ZS-U8 §2.3 PROVEN]. Analogy with biological telomeres: just as telomeres shorten with each cell division, the universe accumulates phase A per cycle, ultimately reaching tunneling (proton decay, etc.). Source: ZS-M3 §7, ZS-F0 §8.3.

Cross-Sector Transduction

The mechanism by which information or coupling constants are transmitted across the X ↔ Y boundary through the Z sector, controlled by the geometric impedance A. All cross-sector interactions carry a factor of A. Source: ZS-M4 (Transduction Principle).

Measure-Projection Weight Theorem (Y/Q = 6/11)

The per-step suppression factor derived from the block-Laplacian log-determinant. Each CP-odd saddle insertion in the path integral contributes a factor $\exp[-\frac{1}{2} \ln(\det L_tot/\det L_Y)] = Y/Q = 6/11$. In the formula $\eta_B = (Y/Q)^{35} = (6/11)^{35}$, the exponent $35 = \mathrm{lcm}(5,7)$ is forced by the Minimality Lemma. Source: ZS-F5 §9 [DERIVED].

Leaky Wilson Loop Identity [PROVEN]

The damping mechanism that makes the i-tetration fixed point z* an attractor rather than divergent. Exact identity for the Lyapunov multiplier: $\lambda^2 = -(\pi^2/4)\cdot(-1)^{z} = -(\pi^2/4)\cdot\exp(i\pi z)$. Therefore $|\lambda^2| = (\pi^2/4)\cdot\eta_topo \approx 0.795 < 1$. The Wilson loop around one X→Z→Y→Z→X cycle "leaks" by a factor $|\lambda^2| < 1$ per cycle, causing convergence to the finite fixed point. Source: ZS-F3 Remark 1.2, ZS-M3.

Mode-Count Collapse

The phenomenon whereby the 1-loop UV structure depends not on individual eigenvalue distributions, but only on the polyhedron's vertex-plus-face count (V+F). For the truncated octahedron: (V+F)/G = (24+14)/12 = 19/6 = a_2(SM) (SM second Casimir) — the unique Archimedean solid with this property. This collapse guarantees that lattice-geometry fine structure does not affect physical predictions, and ensures the spectral density uniformity required by the Measure-Projection Weight Theorem. Source: ZS-S1 [DERIVED]; referenced in ZS-F5 §7.

Tension Overflow (Confinement Mechanism)

Z-Spin's explanation for quark confinement. The "tension" of a state is S + (color load), where S = 0.8915 is the stability threshold. For isolated quarks (color triplet, θ = 120°): Tension = S + √3·A = 1.030 > 1 (exceeds threshold). Color-singlet composites cancel the color load (θ → 0), reducing tension below 1. Source: ZS-A5.

Pinch Identity

The exact identity Pinch(A,θ) = 4 sin(A/2) sin(θ/2), describing the geometric phase accumulated by a color-charged state traversing the seam. For small A: Pinch ≈ 2A sin(θ/2). Source: ZS-A5.

Z-Transformation

The scale-invariant mapping that connects observables across 40 orders of magnitude through the identity $2e^A$, e.g., Ω_Λ/Ω_m = m_d/m_u = $2e^A$ = 2.1668.

Vortex Glass Network

The mechanism for explaining isothermal dark matter profiles in elliptical galaxies. N vortex lines from merger history (each topologically protected by $\pi_1(U(1)) = \mathbb{Z}$) with random S^2 orientations produce $\langle\rho(r)\rangle \propto \ln(r)/r^2$ via exact orientation averaging. Source: ZS-A1 §8.

Z-Bottleneck

The mechanism by which the Z sector (2-dimensional) limits the information-transfer bandwidth between X and Y sectors. Determines the timescale over which quantum coherence is destroyed by gravity. Electromagnetic quantum gate operations (quantum computers) are NOT constrained by the Z-Bottleneck (Area Law non-applicability). Source: ZS-Q1 §2, ZS-F4.

Spinor Phase Gate (Π_Z)

Π_Z(φ_Z) = $\sin^2$(φ_Z/2). Derived from j = 1/2 uniqueness (Theorem 5.1) + SU(2) Wigner d-matrix + physical boundary condition gate(0) = 0. Eliminates the last HYPOTHESIS closure in Z-Sim (ZS-T3). Source: ZS-M3 §12 (v1.0).

Regge-Holonomy Framework

The mathematical foundation of the Z-Telomere mechanism. Phase drift per primitive cell described by Regge curvature: δφ_cell = A·I_cell, I_cell = κ/r = 4/4 = 1. Includes connection to the Immirzi parameter. Source: ZS-M3 §6.

Minimality Lemma

Proves that the minimum number of CP-odd saddle insertions is $n = \mathrm{lcm}(5,7) = 35$, forced by two independent finite-order constraints (pentagonal defect and temporal layer closure). Core component of the $\eta_B = (Y/Q)^{35} = (6/11)^{35}$ formula. Source: ZS-F5 §10.

Dual Z-Anchor

The mechanism deriving the Z-Anchor at the black hole horizon via a dual pathway: Tolman condition + vortex-core condition. Source: ZS-M3.

Z-Bounce

A discrete bounce (reflection) phenomenon corresponding to specific n values in the Z-Spin timescale hierarchy $\tau_n = t_P \times \exp(n\pi/A)$. At $n = \dim(Z) = 2$, it corresponds to weak baryon decays. Source: ZS-S2.

Z-Transit

The non-Abelian holonomy mechanism of CP violation. Phase accumulation along the X→Z→Y→Z→X path generates a CP-odd invariant. Combined with the $\mathrm{lcm}(5,7) = 35$ selection rule. Source: ZS-S6.

Bounded Spurion Framework

The EFT interpretation of seam-breaking. All spurion components must satisfy $|\varepsilon_i| \equiv |\xi_i|/(A \cdot m^*) \leq 1$, ensuring that seam-breaking never exceeds the geometric impedance scale. Source: ZS-S3 Appendix B.

$\mathbb{Z}_2$ Texture Zero Lemma

Proves that when the full EFT respects the μ–τ symmetry (P_{23}), the HNL mass degeneracy is exact at all perturbative orders [PROVEN]. Key to partially resolving ZS-A3's resonance gap (F23-R1). Source: ZS-S4 §3.

Topological Conjugacy

The particle–antiparticle correspondence derived from seam geometry. Connects X-sector protons (winding number $k = 7$) to Y-sector electrons (winding number $k = 1$) through B−L conservation. Source: ZS-S3 §2.

Chamber-Cell Correspondence Lemma

Derives minimum path length $\ell_{min} = 6$ from vertex-link topology: $\Delta^2 \to \mathrm{sd}(\Delta^2) = 6$ chambers → D_3 simply transitive → dual adjacency graph = C_6 → $\mathrm{girth}(C_6) = 6$. Upgraded from HYPOTHESIS to DERIVED-under-P6 in ZS-U8. Source: ZS-U8 (ZS-CH) §3.2a.

X–Z–Y Fractal Symmetry

The structural correspondence between the microscopic triad of forces — Weak (X), Electromagnetic (Z), Strong (Y) — and the macroscopic triad — Dark Matter (X), Gravity (Z), Dark Energy (Y) — sharing an identical algebraic sector assignment via the Lorentz-algebraic decomposition $so(1,3) \cong su(2)_A \oplus su(2)_B$. The mapping $A_k = (J_k + iK_k)/2$ [X-sector] and $B_k = (J_k - iK_k)/2$ [Y-sector] establishes that all six interaction

regimes are scale-separated harmonics of a single Z-Spin effective field theory action. Not self-similarity in the condensed-matter sense, but an algebraic isomorphism between micro and macro physics. Source: ZS-M2 §4, ZS-M4.

Cross-Coupling Theorem

The mathematical theorem that every force formula in Z-Spin involves all three sectors (X, Y, Z) without exception. Proven by enumeration over all six interaction regimes: gravity (X–Z), electromagnetism (Z–Y), weak force (X–Z), strong force (Y–Z), dark matter (X–Z), dark energy (Y–Z) — each necessarily passes through the Z-sector mediator. This proves the universality of A = 35/437 as the cross-coupling impedance. Distinct from cross-coupling in chemistry (Suzuki/Heck reactions). Source: ZS-M2 §5 [DERIVED], ZS-S1.

Axion-Free Strong CP Resolution

The geometric resolution of the strong CP problem without introducing new particles (axions). The Y-sector assignment of SU(3) to $B_k = (J_k - iK_k)/2$ causes the anti-correlated rotation–boost structure to eliminate the spatial orientation required by the θ-term, yielding $\bar{\theta}_{eff} \sim \bar{\theta}_{bare} \times A^2 \times e^{-8\pi^2/g^2} < 5.5 \times 10^{-26}$. Distinct from the Peccei-Quinn/axion mechanism and Nelson-Barr models: here the suppression is a geometric consequence of sector assignment, not a new symmetry or particle. Source: ZS-M2 §6 [DERIVED].

Seam-Charge Projection (Theorem B2)

The theorem establishing that the baryon matter channel corresponds to the $H_X \otimes H_Z$ composite sub-space (dimension X × Z = 6), with the seam projector rank satisfying $\mathrm{rank}(\Pi_{seam}|_{H_X}) = Z = 2$. This yields the baryon density $\Omega_b = XZ/Q^2 = 6/121 = 0.0496$ and the baryon-to-CDM ratio $\Omega_b/\Omega_{cdm} = 6/32 = 3/16 = 0.1875$ (observed: 0.186, 0.8%). Status: DERIVED (A1 resolved via Theorem B3.1, ZS-F5 v1.0: Lorentz algebra route proves A1 without NC-2 dependency). Source: ZS-F5 §6.3, §6.5.

V. Cosmological Quantities

Ω_m^{eff} (Effective Matter Density)

$\Omega_m^{eff} = \Omega_m^{bare} / (1+A) = (38/121)/(1+A) = 0.2908$ [DERIVED — face counting + G_{eff} rescaling]. The BAO-accessible matter density parameter after the G_{eff} rescaling. Distinguished from the bare geometric value $\Omega_m^{bare} = 38/121 = 0.3140$ [DERIVED — face counting, ZS-F2 v1.0 §11]. Source: ZS-U5, ZS-F2 v1.0.

Z-Epoch Coordinate (ν)

The normalized logarithmic time coordinate $\nu(t) = (A/\pi) \ln(t/t_P)$. Current epoch: $\nu(\text{now}) \approx 3.575$. Proton decay: $\nu = 5$. Source: ZS-U5.

Z-Clock

The logarithmic clock of the universe based on the timescale hierarchy $\tau_n = t_P \times \exp(n\pi/A)$. Defined by the Z-Clock coordinate $\nu(t) = (A/\pi) \ln(t/t_P)$. Current $\nu \approx 3.575$, proton decay $\nu = 5$. Step size $C = \pi/A = 39.23$. Source: ZS-M3 §5.

Cosmic Budget (Sector Counting)

The complete set of cosmic density parameters derived from (Z,X,Y) = (2,3,6) and Q = 11: $\Omega_b = 6/121$ (cube faces, DERIVED), $\Omega_c = 32/121$ (truncated icosahedron faces, DERIVED — Boundary Mode Theorem 11.7), $\Omega_m = 38/121 = 0.3140$ [DERIVED — face counting, ZS-F2 v1.0 §11. Cobaya $\Delta\chi^2 = 3.9$ (PASS). Supersedes slot counting 33/121, 39/121 ($\Delta\chi^2 = 226$, FAIL)]. Zero adjustable parameters. Source: ZS-F2 v1.0.

Phase Budget

$\varphi(35) = \varphi(5) \times \varphi(7) = 4 \times 6 = 24$. The Euler totient function of 35 (= 5 × 7, the numerator of A), encoding the spatial ($\varphi(5) = 4 = 2Z$) and temporal ($\varphi(7) = 6 = 2X$) phase structure. Source: ZS-A5.

Capacity (C)

$C \equiv \varphi(35) \times \eta_{topo} = 24 \times 0.3221 = 7.73$ (≈96.6% of 8-layer closure). Testable link: $C \approx k_{L,max} + 1 = 8$ (matter layer closure). Source: ZS-A5.

VI. Black Hole & Proton Decay

Z-Instanton

The non-perturbative tunneling configuration with action $S_{tunnel} = 5\pi/A$, derived from the coset I_h/T_d ($|I_h|/|T_d| = 120/24 = 5$). Predicts proton decay lifetime $\tau_p = t_P \cdot \exp(5\pi/A) \approx 2.56 \times 10^{34}$ years. Source: ZS-A3.

Timescale Hierarchy (τ_n)

$\tau_n = t_P \cdot \exp(n\pi/A)$, producing ~17 orders of magnitude per step. n = 2 matches weak hyperon decays (6.3×10^{-10} s); n = 5 matches proton decay. Step size $\pi/A = 39.225$. Source: ZS-A3.

Sector Duality (BH = X-in-Y)

The structural hypothesis that a black hole is an X-sector structure collapsed into a Y-sector "prison." Inside the Schwarzschild horizon, the r ↔ t exchange maps to X ↔ Y sector exchange. The horizon is the Z-sector boundary ($\varepsilon = 0$). White hole = Y-in-X. Source: ZS-A3.

VII. Particle Physics

33 GeV Ghost

The predicted right-handed Majorana neutrino at $M_R \approx 33.5$ GeV (seesaw with $m_D = m_e \times A = 40.9$ keV). Called "ghost" because its mixing angle $|\theta|^2 \approx 1.5 \times 10^{-12}$ is 10^6–10^7 below current experimental bounds, making it practically invisible at LEP/LHC. Source: ZS-S3.

Three-Generation Limit

$\text{Tension}_\ell(k_G) = S + k_G \cdot A$. With $S = 0.8915$ and $A = 0.0801$: $k_G = 0,1$ stable; $k_G = 2$ borderline; $k_G \geq 3$ forbidden (Tension > 1). Reproduces the three-generation structure of the Standard Model with zero free parameters. Source: ZS-A5.

Molecular Bilinear Program (MBP)

The surviving EWSB candidate mechanism after the CP-instanton pathway was RETRACTED. Uses $\nu = 0$ instanton–anti-instanton (I-$\bar{I}$) molecular sector with $\mu^2_H = (3y^2_t)/(2C_M) \times M^2_P \times \exp(-2S_{cl})$. Currently HYPOTHESIS (strong); independently confirms Spectral VEV via Spectral-Topological Duality. Source: ZS-S4 v1.0.

Mirror Universe / Mirror Cosmology

Introduced in ZS-U8 (ZS-CH). Also referred to as the "mirror sector" in the Z-Spin literature (note: this term is used differently from Foot/Berezhiani mirror matter models). $V_E(-\varepsilon) \equiv V_E(+\varepsilon)$ is an exact algebraic identity, so the mirror sector ($\varepsilon = -1$) is PROVEN observationally identical to the current sector ($\varepsilon = +1$). n_s, r, η_B are all $\mathbb{Z}_2$-symmetric. Decisive near-term test: $r = 0.0089$ (LiteBIRD ~2030, ~6σ distinction from Boyle-Finn-Turok's $r = 0$). Does NOT claim the mirror universe is currently observable (NON-CLAIM). Source: ZS-U8 §5, Book EN §15.5b.

VIII. Ontological Bootstrap (ZS-F0)

Non-Existence Is Self-Contradictory (B0)

The founding meta-logical axiom of Z-Spin. The statement "nothing exists" is a performative contradiction: articulating it presupposes a logical framework, which is "something." In ZF set theory, the empty set $\varnothing$ exists as a mathematical object, and $P(\varnothing) = \{\varnothing\}$ is non-empty. The von Neumann construction $0 = \varnothing$, $1 = \{\varnothing\}$, $2 = \{\varnothing, \{\varnothing\}\}$, … generates all of $\mathbb{N}$ from nothing. Analogous in status to Euclid's parallel postulate or Einstein's light-speed invariance. Source: ZS-F0 §2.1.

Ontological Bootstrap Chain (B0→B8)

B0 (non-existence is self-contradictory) → B1 (Lawvere fixed-point theorem) → B2 (Frobenius extension: $\mathbb{R} \to \mathbb{C}$) → B3 (hyperoperation minimality: tetration) → … → B8. Each step is tagged BOOTSTRAP-PROGRAMME. The Lawvere bridge grounds B0→B1; the Frobenius classification forces the $\mathbb{C}$ extension. Source: ZS-F0. [v1.0: ZS-F0 v1.0(Revised), April 2026 — the BOOTSTRAP-PROGRAMME tag now applies only to B0 itself. All four ontological links B0→B1 (Lawvere, DERIVED), B1→B2 (Frobenius,

DERIVED-CONDITIONAL Level A), B2→B3 (hyperoperation minimality, DERIVED), and the Möbius–seam $W^2 = I$ correspondence (DERIVED-CONDITIONAL) are closed via the Stage 1–5 closure program. All nine F-BOOT gates are now closed or passed (6/9 DERIVED, 2/9 PASS, 1/9 DERIVED + TESTABLE). The chain from B0 to A = 35/437 is therefore DERIVED conditional only on the meta-logical founding axiom B0. The Fundamental Fixed Point Principle (FFPP, ZS-F0 v1.0(Revised) §13) compresses all Z-Spin numerical content under a minimal five-axiom set to the single transcendental complex value $W_0(-i\pi/2)$. See Chapter 0.3 and §29.11 annotations for full closure summary.]

IX. Methodology & Epistemic Framework

Epistemic Status System

Z-Spin's mandatory classification hierarchy for all claims: PROVEN → DERIVED → DERIVED-CONDITIONAL → HYPOTHESIS → OBSERVATION → BOOTSTRAP-PROGRAMME → NON-CLAIM → RETRACTED. Every result must carry an explicit status label.

Anti-Numerology Discipline

The non-negotiable requirement that every numerical result must have a complete geometric or standard-physics derivation chain. Validated by Monte Carlo scans (typically 500K+ random samples) to confirm non-coincidence. Ordering principle: compute first, compare second — never reverse.

Falsification Gate (F-numbered)

Pre-registered, quantitative conditions under which specific Z-Spin predictions would be falsified. Each gate has an ID (e.g., F-A3.7), a kill condition, the relevant experiment, and a timeline. Falsification conditions must be stated before computing results.

NON-CLAIM

An explicit declaration that a quantity is NOT derived from the framework. Used for intellectual honesty — e.g., the Higgs self-coupling ($\lambda_H \approx 0.129$) remains a NON-CLAIM. The pattern "120+5 = 125" was explicitly rejected as numerology for the Higgs mass. [v1.0: The Higgs VEV, formerly the canonical NON-CLAIM example, was upgraded to DERIVED in v1.0/ZS-S4 v1.0.]

BOOTSTRAP-PROGRAMME

Epistemic status for the ontological-to-mathematical bridge (B0→B8 chain). Self-consistent and falsifiable, but cannot be proven within the framework itself — analogous to axiom status in formal mathematics. Source: ZS-F0.

X. Paper System & Organization

5-Theme v1.0 System

The organizational structure of Z-Spin papers: ZS-F (Foundations, F0–F6), ZS-M (Mathematical Spine, M1–M12), ZS-S (Standard Model Completion, S1–S7), ZS-U (Early Universe, U1–U5), ZS-A (Astrophysics, A1–A5).

The Book (EN)

The companion English monograph synthesizing the full Z-Spin framework across all papers. Version-controlled in parallel with source papers. Currently v1.0.

Sogaeseo (Korean Introduction)

The Korean-language introduction document summarizing Z-Spin Cosmology for a general audience. Maintained in parallel with the Book EN.

XI. Polyhedral Geometry

Polyhedral Duality

The connection between Platonic solid dual pairs and the X/Y/Z sector structure. Z: self-dual tetrahedron pair. X: cube ↔ octahedron (mediated by truncated octahedron). Y: dodecahedron ↔ icosahedron (mediated by truncated icosahedron). Source: ZS-F2.

O_h × I_h Defect Structure

The combined octahedral (O_h, order 48) and icosahedral (I_h, order 120) symmetry groups whose polyhedral defect geometry fixes A = 35/437. Source: ZS-F2.

Truncated Octahedron Lattice (Kelvin Cell)

The unique Archimedean solid satisfying $(V+F)/G = a_2(SM)$ (the second Casimir of the SM gauge group). 24 vertices, 36 edges, 14 faces. Also known as the Kelvin Cell — the Voronoi cell of the BCC lattice. Requires 72 qubits for SU(2) lattice simulation — feasible on current quantum hardware. Source: ZS-A4, ZS-Q1 §2.1.

XII. Quantum Computing & Hardware (ZS-QC)

Z-Sim

The quantum computer engine simulating Z-Spin physics. Has 8 closure parameters; in v2.2 (March 2026), the Spinor Phase Gate (Π_Z) eliminated the last HYPOTHESIS closure, making Z-Sim fully DERIVED. Source: ZS-T3.

X-Material / Y-Material / Z-Material

Three physical material categories for Z-Spin quantum computer hardware. X-Material: computational register. Y-Material: 6-channel dissipative reservoir. Z-Material: 2-channel topologically protected mediator. Parasitic X–Y coupling must be below 1%. Source: ZS-QC (ZS-Q6).

Z-ON / Z-OFF

Activation/deactivation states of the Z-Material. Z-ON: normal X↔Y mediation via Z-path. Z-OFF: Z-Material detuned by >10Γ. $g_XY = |S_{31}(Z\text{-}OFF)|/|S_{31}(Z\text{-}ON)| < 0.01$ is the PASS condition. Source: ZS-QC.

Kill-Switch (KS)

Operational abort switches for the Z-Spin quantum computer with concrete measurement protocols. KS-2: Z-Seam Collapse (if u_seam = O(1), Z is non-functional). KS-3: Direct Path Z-OFF (if direct path without Z exceeds noise floor, architecture is rejected). Source: ZS-QC.

Seam Bus

The inter-cell Z-seam bus. Communication pathway via Z–Z coupling between cells in multi-cell quantum computers. Source: ZS-QC.

XIII. Cyclic Cosmology

Z-Telomere Cyclic Cosmology

Cyclic cosmology based on the Z-Telomere mechanism. Because $V_E(+1) = V_E(-1) = 0$ exactly ($\Delta V = 0$), Coleman–de Luccia tunneling is strictly forbidden ($S_CdL \to \infty$) [PROVEN, ZS-U8 §2.3]. The transition is instead a topological holonomy winding-change: phase accumulation $\delta\varphi = A$ per Regge cycle reaches a unit winding (2π) after $N_{2\pi} = 2\pi/A \approx 78.45$ cycles, and the Z_2 vacuum transition ε: $+1 \to -1$ proceeds via the D_3 holonomy action $S_Z_2 = 6\pi/A \approx 235.35$. $\tau_6/\tau_5 = \exp(\pi/A) \approx 10^{17}$. Source: ZS-U8 §2–§3, ZS-A6 Corollary II, ZS-F0 §8.3.

Z-Telomere Collapse

The specific phase-transition event triggered when the accumulated Z-Telomere phase drift reaches the holonomy threshold $S_tunnel = 5\pi/A \approx 196.13$. The non-minimal coupling $(1+A\varepsilon^2)R$ generates a phase drift $\delta\varphi = A \approx 0.0801$ radians per Regge cycle near topological defects; after sufficient accumulation, a unit winding-number change is triggered on the field configuration space, and the ε-field undergoes a topological holonomy transition from the $\varepsilon = +1$ vacuum to the $\varepsilon = -1$ mirror vacuum. This is not a Coleman–de Luccia bounce: because $\Delta V = V_E(+1) - V_E(-1) = 0$ exactly, $S_CdL \to \infty$ and CdL tunneling is strictly forbidden [PROVEN, ZS-U8 §2.3, ZS-A6 §5.1]. The transition is uphill in V_E and driven by holonomy accumulation, not potential-energy barrier penetration. This is the Z-Spin analog of biological telomere shortening triggering cellular senescence. Source: ZS-U5 §5, ZS-M3 §7, ZS-U8 §2–§3, ZS-A6 Corollary II.

Post-τ_5 Dynamics

XIV. Phase 5 Closing Papers Terminology [v1.0: Phase 5 Addendum, April 2026]

Gap G2 (Closure)

The assignment of icosahedral group $I \cong A_5$ irreducible representations to Standard Model field classes given by ZS-M9 Table 2 (originally HYPOTHESIS strong, ZS-U9 §8.2). CLOSED at DERIVED-strong status in the Phase 5 cycle via two independent routes: Route (b) Falsification + Z5-McKay Handedness Theorem (ZS-M15 §8.4l, DERIVED) and Route (a) Action-Level Factorized Spectral Determinant (ZS-M16 §8.4m, DERIVED). The path-integral arm of NC-M16.1 is further closed by ZS-M17 §15.9 via OS reconstruction. Full PROVEN status still requires non-perturbative lattice Monte Carlo (TESTABLE on IBM Eagle 127q hardware). Source: ZS-M15, ZS-M16, ZS-M17 v1.0.

Gauge-Yukawa Spectral Relation (★)

The action-level relation $g^2_2 \cdot C_0^2 = (d_eff/Z) \cdot y_t^2 \cdot C_M \cdot \exp(2\delta)$ connecting the 30-3 closure formula and the Molecular Bilinear Program (MBP) formula for the Higgs quartic coupling λ_H. CLOSED at DERIVED status in the Phase 5 cycle via the four-identity proof of ZS-S13 §8.4n: $|O_h| = X \cdot C_0$, the triple identification $b_1 = X = N_c = 3$, the $X + Y = X^2 = d_eff$ identity (forced by $Y = X(X - 1)$ for $X = 3$), and the new Cheeger-Müller-type decomposition $C_M = C_M^{sp} + X \cdot \ln G$. Yields the closed-form prediction $m_t = 171.872$ GeV (TESTABLE; FCC-ee decisive ~2040s) using zero observed Higgs-sector inputs. Source: ZS-S13 v1.0; companion to ZS-S4 §6.16.

Cheeger-Müller-type Decomposition

The structural identity $C_M = C_M^{sp} + X \cdot \ln G$ decomposing the full BCC T^3 spectral invariant into a coexact (Y-sector) piece C_M^{sp} and a harmonic+exact (X-sector) piece $X \cdot \ln G = b_1 \cdot \ln G$. Numerically: $16.17795 = 8.72323 + 3 \times 2.48491 = 8.72323 + 7.45472$ (exact at 50-digit mpmath precision). Registered for the first time in ZS-S13 §3.4 / §8.4n as a clean structural theorem. Source: ZS-S13 v1.0 Identity (iv).

Z5-McKay Handedness Theorem

The structural theorem (ZS-M15 §5, DERIVED) that breaks the residual $3 \leftrightarrow 3'$ ambiguity in the I-irrep-to-SM-field-class assignment. Uses the McKay correspondence chain $Z_5 \subset SU(2) \to \hat{A}_4 \to SU(5)$ (ZS-M9 §5 DERIVED) plus the Z_5 charge complementarity $\{\omega^1, \omega^4\} \subset 3{\downarrow}Z_5$ vs $\{\omega^2, \omega^3\} \subset 3'{\downarrow}Z_5$ to identify I-irrep 3 with the SU(2)_L simple root ω^4 and I-irrep 3' with the U(1)_Y simple root ω^3. Under SM left/right handedness, this forces LH ↔ I-3 and RH ↔ I-3'. Source: ZS-M15 v1.0.

Factorized Spectral Determinant Order Parameter ΔΓ_G2

The Route (a) order parameter $\Delta\Gamma_G2 = \gamma_R \cdot C_G2^{sp} / 2 = -5.2030934754304919584540850007452415006579790712319$ (50-digit mpmath precision), derived from the chirality-graded supertrace of the Y-sector Hodge-Dirac operator D_TI on the I-equivariant Schur projection onto I-irrep 3 vs I-irrep 3'. The sign $\Delta\Gamma_G2 < 0$ selects Assignment A (LH ↔ I-3, RH ↔ I-3', gauge ↔ I-4, Higgs ↔ I-5, ν_R ↔ I-1), structurally protected by the D_5 harmonic decomposition. Source: ZS-M16 v1.0.

OS Reconstruction (Z-Spin)

The Osterwalder-Schrader reconstruction of the Z-Spin continuum quantum field theory from the reflection-positive Wick-rotated lattice action. Established in ZS-M17 §15.9 via Theorem M17.7, combining operator-norm convergence (M17.1), reflection positivity (M17.3), and universality (M17.6) into the standard OS reconstruction theorem. CLOSES OP-c.1 (continuum limit rigor) of ZS-Q5 §8.6 at DERIVED status. The Wick-rotated Z-Spin action satisfies OS positivity because $A > 0$ and $\varepsilon^2 \geq 0$ ensure $(1 + A\varepsilon^2)R$ is positive definite. Source: ZS-M17 v1.0.

Photon Pillar III (E ⊥ B from V_XZ ⊥ V_ZY Complex-Conjugate Pairing)

The HYPOTHESIS strong proposal of ZS-S12 §5.13 that the 90° orthogonality between electric field E and magnetic field B in any propagating electromagnetic plane wave is the 4D spacetime projection of the complex-conjugate pairing $V_{ZY} = (V_{XZ})^*$ between the two Z-sector mediation channels. The phase difference $\arg(V_{XZ}) - \arg(V_{ZY}) = \theta$ is fixed by the half-angle SU(2) spinor structure of the Z-sector. Falsification path: precision polarization interferometry sensitive to $V_{ZY} - (V_{XZ})^*$ deviations at machine precision. Source: ZS-S12 v1.0.

V1 (Planck-to-Bulk Handoff Closure)

The matching question of ZS-M12 §7.6: does the U(1) comoving Noether charge $Q = a^3\varepsilon^2\dot{\theta}$ survive the ~3 τ_P dissipative Auto-Surgery phase? CLOSED at DERIVED-CONDITIONAL status in the Phase 5 cycle by ZS-U11 §15.5e via four independent U(1) charge-protection channels: Channel 1 — U(1) Noether anomaly-freeness [PROVEN]; Channel 2 — Z_2 seam topological winding preservation [DERIVED]; Channel 3 — z* damped spiral acts on radial mode only [DERIVED]; Channel 4 — Quantum Foam Engine equipartition [HYPOTHESIS strong]. Conditional on (i) preservation of U(1) by quantum gravity at the A scale (F-M12.4); (ii) Quantum Foam Engine equipartition (NC-M12.3). Source: ZS-U11 v1.0.

Multi-Channel U(1) Protection

The four-channel framework of ZS-U11 §15.5e for analyzing the survival of the U(1) Noether charge Q across distinct decay pathways. Each channel acts at a structurally different level of the framework (field theory, configuration topology, mode separation, statistical ensemble), so a single channel failure does not automatically propagate to the others. Provides the structural basis for the V1 OPEN → DERIVED-CONDITIONAL upgrade.

Closing Papers Cycle

The seven April 2026 papers (ZS-S11, ZS-M15, ZS-M16, ZS-M17, ZS-S12, ZS-S13, ZS-U11) integrated as Phase 5 of the v1.0 Book integration program, completing the corpus at 70 papers. Distinct from the Phase 4 Bridge Papers cycle (ZS-M14, ZS-S9, ZS-S10, ZS-U10) which integrated the four electron-related synthesis/closure papers. The Closing Papers achieve: SM anomaly cancellation reformulation (S11), two-route Gap G2 closure (M15 + M16), continuum-limit rigor + OS reconstruction (M17), photon synthesis (S12), Higgs branch unification + m_t = 171.872 GeV (S13), V1 closure (U11). Capstone paper: ZS-U11 (Paper 70 of 70).

Phase 5 Cascade Decision (April 23, 2026)

The user editorial decision of April 23, 2026 to apply downstream cascade status changes uniformly across the body and appendices of The Book in this Phase 5 cycle, replacing prior dated-note-only conventions on the specific cascades arising from Phase 5 closures (Gap G2, Trinity Braiding Theorem, ZS-S4 §6.8/§6.11/§6.16, OP-c.1/c.2/c.3, V1). All pre-Phase-5 historical content is otherwise preserved verbatim per the no-deletion rule. External label remains v1.0 (no version bump).

The evolution of the universe after proton decay (τ_5). Hubble rate $H_{dS} \approx 1.18 \times 10^{-61}\, M_P$ (Planck 2018 Λ). $m_{eff}/H_{dS} \sim 10^{60}$ ensures $\varepsilon = +1$ is pinned. Source: ZS-U8 §6.

This glossary covers terms identified by systematic grep search across all 10 project files containing 59 papers (ZS-F0–F8, ZS-M1–M13, ZS-S1–S8, ZS-U1–U9, ZS-A1–A7, ZS-Q1–Q7, ZS-QH/QS/QC, ZS-T1-T3) plus the Book EN v1.0. Total: 88 entries across 13 categories.

XV. Phase 7 Terminology [v1.0 dated 2026-04-24]

Terms introduced or formalized in the Phase 7 Post-Closure Addendum integration of ZS-M18, ZS-A8 v1.0 Revised, and the ZS-U8 v1.0 dated update 2026-04-24. All terms inherit their epistemic status from the source papers; no new framework axioms are introduced.

Y-Time Dilation Theorem The X-clock observation factor $\exp(\pi/A) \approx 1.08\times10^{17}$ by which any Y-sector completion event appears dilated relative to the Y-sector proper-time measurement of the same event. Decomposes structurally as $\exp((2\pi/A) \times (1/2))$ where $N_{(2\pi)} = 2\pi/A$ is the Z-Telomere completion cycle count (PROVEN, ZS-U5) and $\langle\sin^2(\varphi/2)\rangle = 1/2$ is the SU(2) spinor phase gate time-average over the 4π period (PROVEN, ZS-T2 §5.5). Resolves the 10^{17} cyclic-cosmology inefficiency puzzle. [STATUS: HYPOTHESIS-strong] Source: ZS-A8 v1.0 Revised §5.3 Theorem 5.3.1; ZS-U8 v1.0 dated 2026-04-24 §4.1; Book §15.5f.2.

Parallel Reading (Dual-Clock Reading) The dual-clock interpretation of the timescale hierarchy $\tau_n = t_P \times \exp(n\pi/A)$ under which the X-sector baryon-completion lifecycle (τ_5) and the Y-sector wave-contraction lifecycle (Z_2 holonomy completion) are SIMULTANEOUS in their respective sectoral proper times, not sequential. The 10^{17} ratio τ_6/τ_5 is reinterpreted as the Y-sector time-dilation seen by the X-clock observer rather than a temporal waiting interval. Contrast: the SEQUENTIAL READING (sequential X-clock reading) interprets the same hierarchy as a single ordered list of events in the X-observer proper time, with τ_6 occurring 10^{17} baryon-decay-times after τ_5. The two readings are mutually consistent with all numerical predictions; both are preserved in the Book per the no-deletion rule. Source: ZS-U8 §4.1; Book §15.5a Phase 7 Note and §15.5f.

Wave-Contraction Sector The Y-sector dynamics governing wave-to-particle transition at Planck scale, as the SYMMETRIC counterpart to the X-sector expansion dynamics. Characterized by the contraction scale $Y^2(1-2A) = 30.23$ (Bridge 2 of ZS-A8 §4), where $Y^2 = 36 = X\cdot Z\cdot Y = E$(truncated octahedron) is the wave-channel scale (PROVEN, ZS-F7 §4.4) and $(1-2A)$ is the leading Taylor expansion of $1/(1+A)^2$. The wave-contraction dynamics are derived consequences of the existing $(1+A\varepsilon^2)R$ action (NC-A8.5: ZS-A8 does NOT propose a new physical action). [STATUS: DERIVED] Source: ZS-A8 v1.0 Revised §1.2; Book §15.5f.3.

Expansion-Contraction Symmetry Theorem 6.1 of ZS-A8 v1.0 Revised: every Z-Spin expansion phenomenon governed by $(1+A)$ has a contraction-side counterpart governed by $(1-2A)$ = LO Taylor of $1/(1+A)^2$. Both are determined by the same geometric impedance $A = 35/437$. Three-facet manifestation: (i) EXPANSION as $\exp(A) = 1.0834$ (Hubble ratio, X-sector); (ii) CONTRACTION as $Y^2(1-2A) = 30.23$ (wave-channel scale, Y-sector); (iii) TIME-DILATION as $\exp(\pi/A) = 1.08\times10^{17}$ (Y-completion observation factor). The algebraic identity $(\exp A)^{(\pi/A^2)} = \exp(\pi/A)$ confirms a single exponential family indexed by powers of A. Numerical symmetry: $(1+A)\cdot(1-2A) = 0.9071$. [STATUS: DERIVED] Source: ZS-A8 v1.0 Revised §6 Theorem 6.1; Book §15.5f.3.

Polyhedral-Tetration Bridges (Bridge 1, Bridge 2, Bridge 3) Three formulas of ZS-A8 v1.0 Revised connecting the polyhedral defects ($\delta_X = 5/19$, $\delta_Y = 7/23$, both PROVEN in ZS-F2) to the i-tetration fixed point $z^* = 0.4383 + 0.3606i$ (PROVEN in ZS-M1). Bridge 1 [OBSERVATION-strong]: $|z^*| \approx \delta_X + \delta_Y = 248/437$ at 0.0087% relative accuracy. Bridge 2 [OBSERVATION-strong]: $\eta_topo \approx B^2 + (\delta_Y - \delta_X)^2 / [Y^2(1-2A)]$ at 0.000001% relative accuracy. Bridge 3 [DERIVED-CONDITIONAL]: $\Delta = 1/2 - x^*$ reconstructed from polyhedral inputs via Locking L2 self-consistency at 0.000002% via iteration. Anti-numerology validation: 500,000-sample three-basket Monte Carlo confirms structural uniqueness (p_trial = 0.0048%, STRONG PASS). Source: ZS-A8 v1.0 Revised §3, §4, §5; Book §15.5f.4.

Symmetry-Asymmetry Unified View (Tilt-from-Balance Z-Anchor) Section SA of ZS-A8 v1.0 Revised, providing a Z-anchor that consolidates thirteen corpus-wide connections under one principle: dynamics is the tilt from balance. Three numbers play complementary roles: unity (1) is the self-identity of the universe; (1/2) is the unique fixed point of the Z_2 involution $x \leftrightarrow 1-x$ and is the natural balance point of any Z_2-symmetric system; A = 35/437 is the joint tilt of the universe, the product $\delta_X \cdot \delta_Y$ of two sectoral asymmetries (PROVEN, ZS-F2). The X-frame describes the universe as accelerating expansion; the Y-frame describes the same universe as decelerating contraction. Frame-invariance of A: the geometric impedance has the same value 35/437 in both frames; only the observable manifestation changes. [STATUS: HYPOTHESIS-strong (INTERPRETATION)] Promotion path: formalize the information-time correspondence as a theorem unifying ZS-Q6 §5 (dim(Z) = 2 tensor-network bond dimension), ZS-A3 (Bekenstein horizon entropy), and ZS-Q7 §6 (Z-mediator channel-capacity). Source: ZS-A8 v1.0 Revised §SA; Book §15.5f.5.

RH-Inclusive Reading of Z-Spin Entry H21 of ZS-M18 v1.0 dated 2026-04-24, tagged DERIVED-interpretation, synthesizing four PROVEN/DERIVED corpus elements (ZS-M13 §2.2 Eisenstein–Dedekind factorization $\zeta_{\mathbb{Q}(\omega)}(s) = \zeta(s) \cdot L(s, \chi_{-3})$; ZS-M18 H11 $\sigma = 1/2 \leftrightarrow j = 1/2$ isomorphism; ZS-M1 HSI conformality at z*; ZS-A8 §6 + §5.3) without introducing any new prediction or parameter. Positions Z-Spin as a unified frame embracing the Riemann Hypothesis without requiring its proof: the $\sigma = 1/2$ critical line and the Z_2 reflection fixed point at $x = 1/2$ are two faces of the same dim(Z) = 2 involution. Falsification gate F-M18.11 OPEN, awaits eventual RH resolution. NC-M18.3 preserved: this reading does NOT claim a proof of the Riemann Hypothesis. Source: ZS-M18 v1.0 dated 2026-04-24 H21; Book §15.5f.6.

Glossary scope update [Phase 7] Phase 7 Glossary scope update: the original Glossary v1.0.0 scope statement of 'systematic grep search across all 10 project files containing 59 papers (ZS-F0–F8, ZS-M1–M13, ZS-S1–S8, ZS-U1–U9, ZS-A1–A7, ZS-Q1–Q7, ZS-QH/QS/QC, ZS-T1-T3) plus the Book EN v1.0' is preserved verbatim per the no-deletion rule. The current corpus scope at the Phase 7 Post-Closure Addendum is 73 papers (Phase 5 closure: 70 + Phase 6: 1 (ZS-S14) + Phase 7: 2 (ZS-M18 + ZS-A8 v1.0 Revised)) plus the Book EN v1.0. Total Glossary entries after Phase 7 addition: 88 + 7 (Category XV) = 95 entries across 14 categories.